

fn match_group_by_truncation

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This proof resides in “contrib” because it has not completed the vetting process.

Proves soundness of `match_group_by_truncation` in `mod.rs` at commit `f5bb719` (outdated¹).

1 Hoare Triple

Precondition

Compiler-verified

Types matching pseudocode.

Caller-verified

None

Pseudocode

```
1 def match_group_by_truncation(  
2     plan: DslPlan, identifier: Expr  
3 ) -> Optional[Tuple[DslPlan, Truncation, Bound]]:  
4     if not isinstance(plan, DslPlan.GroupBy): #  
5         return None  
6  
7     input = plan.input  
8     keys = plan.keys  
9     aggs = plan.aggs  
10    apply = plan.apply  
11    options = plan.options  
12  
13    if apply is not None or options != GroupbyOptions.default():  
14        return None #  
15  
16    ids, by = partition(  
17        lambda expr: expr == identifier, keys  
18    ) #  
19  
20    if not ids: #  
21        return None  
22  
23    return (  
24        input,
```

¹See new changes with `git diff f5bb719..0026d99 rust/src/transformations/make_stable_lazyframe/truncate/matching/mod.rs`

```

25         Truncation.GroupBy(keys, aggs),
26         Bound(by=by, per_group=1, num_groups=None), #
27     )
28
29
30 # part of Rust's standard lib, included for readability, with hardcoded types
31 def partition(
32     predicate, iterable: Iterable[Expr]
33 ) -> Tuple[HashSet[Expr], HashSet[Expr]]:
34     true_set = set()
35     false_set = set()
36     for item in iterable:
37         if predicate(item):
38             true_set.add(item)
39         else:
40             false_set.add(item)
41     return true_set, false_set

```

Postcondition

Theorem 1.1 (Postcondition). For a given query plan and user identifier expression, if the query plan bounds row contributions per-identifier via a group-by, returns a triple containing the input to the truncation, the truncation itself, and the per-id bound on user contribution.

Proof. Lines 4-14 check if the query plan is a simple group-by, and exits without a match if not.

A valid group-by truncation is a group-by where the grouping keys contain a user identifier. So line 18 splits the group-by keys into two sets, and line 20 rejects the match if the user identifier is not in the group-by keys.

The remaining grouping columns are then reflected in the bound on user contributions, on line 26. $\square$